ClinIQ: Agentic Clinical Pattern Recognition and Decision Support Platform

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Cotiviti Internship Assessment

July 2026

Introduction

Clinical decision making and pattern recognition in healthcare involve using data, rules, machine learning, and generative AI to identify meaningful relationships between diagnoses, procedures, documentation, policies, and review outcomes. Clinical decision support systems are designed to provide relevant information that helps users make or review healthcare decisions (Agency for Healthcare Research and Quality [AHRQ], n.d.). For Cotiviti, this topic is relevant because payment accuracy and claim review require both clinical understanding and operational judgment. This report explores how agentic clinical decision intelligence can support evidence-aware healthcare claim review. As a proof of concept, ClinIQ demonstrates one possible implementation of this concept.

Topic Concept and Trends

Agentic clinical decision intelligence combines generative AI, retrieval-augmented generation, pattern recognition, and human-in-the-loop decision support. In claim review, such a system can analyze diagnosis codes, procedure codes, supporting documents, payer information, provider details, and authorization status, then validate the case, retrieve evidence, identify verification gaps, and generate reviewer-facing guidance.

This approach reflects current healthcare AI trends. Organizations are using AI to reduce evidence-gathering effort and organize complex documentation. Retrieval-augmented generation helps ground healthcare AI outputs in evidence instead of unsupported model responses. Human-in-the-loop workflows are important because coverage, medical necessity, and payment decisions require transparency and reviewability. For Cotiviti, these trends align with public-facing payment accuracy and claim review priorities (Cotiviti, n.d.). The topic also connects to classification, prediction, inference, clustering, and time-series anomaly detection. These methods can categorize evidence, estimate review risk, identify verification gaps, group similar claim patterns, and support future analysis of repeated procedures or unusual provider behavior.

Opportunities and Threats

Agentic decision-support systems create opportunities for reviewer efficiency, payment integrity, and explainability. Reviewers could receive a structured summary of claim details, detected patterns, retrieved evidence, verification requirements, and suggested actions. This can reduce manual searching and standardize evidence review. Cotiviti's payment accuracy solutions focus on improving payment integrity across claim review workflows, making evidence-aware support strategically relevant (Cotiviti, n.d.).

The main risk is over-reliance on AI. If a system appears to make a final claim decision, reviewers may treat the output as approval or denial even when important internal information is missing. Recent health policy analysis notes that AI use in prior authorization and claims review raises consumer protection concerns when human judgment, appeals, or coverage decisions may be affected (KFF, 2026). A responsible approach should avoid approve or deny language and use terms such as Recommended Next Action and Internal Verification Required.

Another risk is evidence quality. Online retrieval can return useful sources, but it can also return weak, outdated, or unrelated information. Public sources may support clinical, coding, and medical necessity reasoning, but they cannot reliably verify payer-specific benefits, provider network status, authorization records, or member eligibility. Coverage decisions still depend on whether services meet coverage and medical necessity requirements, so public evidence should not be treated as final claim authority (Centers for Medicare & Medicaid Services [CMS], 2026).

Strategic Recommendation

Cotiviti could explore three strategic actions: build an internal reviewer-assistance layer, connect AI retrieval pipelines to verified payer and policy repositories, and position human-in-the-loop decision intelligence as a responsible alternative to automated claim adjudication. A system like ClinIQ could connect structured claim data with pattern recognition, evidence retrieval, verification awareness, and explainable recommendations.

As a proof of concept, ClinIQ demonstrates this strategy by analyzing structured claim JSON, retrieving public evidence, identifying internal verification gaps, and generating a decision-support report. The platform intentionally keeps final authority with human reviewers. This makes the project a responsible AI approach: it improves evidence organization and reviewer guidance without pretending that public evidence is enough for final claim adjudication.

Conclusion

Agentic clinical decision intelligence and pattern recognition offer Cotiviti a practical opportunity to improve healthcare claim review. The strongest near-term value is not automated claim approval, but transparent reviewer support that organizes evidence, exposes uncertainty, and recommends next actions while preserving human oversight.

References

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